

Figure 1. D1 sub-score → grade reference matrix (DO/ORP-only main link, v1.1)

| criterion            |                       | A ( $\geq 4.5$ ) | B (3.5–4.5) | C (2.5–3.5) | D (1.5–2.5) | F ( $< 1.5$ )              |
|----------------------|-----------------------|------------------|-------------|-------------|-------------|----------------------------|
| Q_spike<br>(spike)   | spike rate $\leq 2\%$ | very rare        | rare        | occasional  | frequent    | very frequent ( $> 20\%$ ) |
| Q_step<br>(step)     | KS statistic          | near 0           | low         | moderate    | elevated    | high (sustained)           |
| Q_drift<br>(drift)   | PLS residual z        | $ z  < 1.5$      | 1.5–2.0     | 2.0–2.5     | 2.5–3.0     | $> 3.0$ sustained          |
| Q_freeze<br>(freeze) | RLE / unique ratio    | no freeze        | $< 5$ min   | 5–15 min    | 15–60 min   | $> 60$ min                 |
| Q_regime<br>(regime) | W1 normalised         | $< 1.0$          | 1.0–2.0     | 2.0–3.0     | 3.0–4.0     | $> 4.0$                    |